

Signatures of Microevolutionary Processes in Phylogenetic Patterns

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Abstract.—Phylogenetic trees are representations of evolutionary relationships among species and contain signatures of the processes responsible for the speciation events they display. Inferring processes from tree properties, however, is challenging. To address this problem, we analyzed a spatially-explicit model of speciation where genome size and mating range can be controlled. We simulated parapatric and sympatric (narrow and wide mating range, respectively) radiations and constructed their phylogenetic trees, computing structural properties such as tree balance and speed of diversification. We showed that parapatric and sympatric speciation are well separated by these structural tree properties. Balanced trees with constant rates of diversification only originate in sympatry and genome size affected both the balance and the speed of diversification of the simulated trees. Comparison with empirical data showed that most of the evolutionary radiations considered to have developed in parapatry or sympatry are in good agreement with model predictions. Even though additional forces other than spatial restriction of gene flow, genome size, and genetic incompatibilities, do play a role in the evolution of species formation, the microevolutionary processes modeled here capture signatures of the diversification pattern of evolutionary radiations, regarding the symmetry and speed of diversification of lineages. [Diversification rates; evolutionary radiation; individual-based model; parapatry; speciation; sympatry.]

Understanding the origin and maintenance of species diversity is a major goal in ecology and evolutionary biology. Processes influencing speciation have been studied in numerous ways, testing how different ecological and non-ecological processes contribute to the generation of species (Nee et al. 1992; Pybus and Harvey 2000; Rundell and Price 2009; Butlin et al. 2012; Gavrillets 2014; Morlon 2014; Seehausen et al. 2014). Model-based approaches have played an essential role in this avenue (Gavrillets 2014) and, over the past years, have moved from providing proofs of concept of novel processes (Dieckmann and Doebeli 1999; de Aguiar et al. 2009; Servedio et al. 2014; Melián et al. 2015) to tools which generate predictions that can be directly contrasted with empirical patterns (McPeck 2008; Gavrillets and Losos 2009; Pigot et al. 2010; Martins et al. 2013; Gascuel et al. 2015; Manceau et al. 2015).

An important way to unveil, which processes promote speciation is to identify their fingerprints in the macroevolutionary patterns of phylogenetic trees (Nee et al. 1992; Morlon 2014). Model-based approaches derived from the Neutral Theory of Biodiversity (NTB) have been successful at predicting empirical macroecological patterns such as species abundance distribution and species-area relationship (Etienne and Alonso 2005; Jabot and Chave 2009; O’Dwyer and Green 2010; Hurlbert and Stegen 2014), but have not provided accurate predictions about macroevolutionary patterns such as phylogenetic tree shape. Most of these are birth–death models that consider speciation (birth) and extinction (death) as random events, not

taking into account microevolutionary processes, as the population dynamics within species, which could explain their deficiency to predict macroevolutionary patterns (Mooers et al. 2007; Davies et al. 2011). Speciation models that incorporate microevolutionary processes along with phylogenetic trees arising from the theoretical simulated populations are key to clarify how population-level processes affect diversification rates and the emerging macroevolutionary patterns, filling the micro-macro gap of the biodiversity theory. These models are rarely utilized because of their complexity, but they describe individual-level processes that can directly scale-up to influence diversification (Rosindell et al. 2015).

The interest in searching for signatures of microevolutionary processes in phylogenetic trees using mechanistic simulation models has increased over the past years (Cabral et al. 2017). Previous studies have explored the phylogenetic signatures of speciation processes that evolves by accumulation of incompatibilities in the context of demographical non-equilibrium in sympatry (Manceau et al. 2015), or of ecological spatial heterogeneity in metacommunities (Gascuel et al. 2015). These studies provide interesting insights about microevolutionary process driving speciation and their emerging phylogenetic patterns, nevertheless, the role of geographical isolation and other population level ingredients, such as population density, dispersal and number of genes, on the build-up of reproductive isolation have not been investigated in the context of the NTB. Different macroevolutionary

patterns, such as branching slowdown, have been attributed to ecological modes of speciation (Gavrilets and Vose 2005; Gascuel et al. 2015). However, these patterns can also be predicted by non-ecological speciation processes (Pigot et al. 2010; Wang et al. 2013). Studies focusing primarily on non-ecological processes seldom incorporate microevolutionary processes (but see (Manceau et al. 2015)), hence the importance of filling such a gap in the literature.

Here, we link microevolutionary processes such as gene flow and the evolution of reproductive isolation, to macroevolutionary patterns displayed by phylogenetic trees. More specifically, we identify the signatures that different geographical modes of speciation and genome size leave on phylogenetic patterns. We use a spatially explicit individual-based model (IBM) of speciation in which reproductively isolated species evolve in response to genetic and spatial restrictions imposed on reproduction (de Aguiar 2017). The model has the potential to explain macroecological patterns compatible with the NTB (de Aguiar et al. 2009), patterns of genetic differentiation between species (Martins et al. 2013) and diversification on a macroevolutionary scale (Costa et al. 2017). Our approach differs from previous studies in two important aspects: first, we explore the connection between population genetics, speciation theory and macroevolution by studying the phylogenetic patterns of a mechanistic model for the evolution of reproductive isolation that explicitly simulates sexual reproduction, dispersal, mutation and genetic drift. We model genomes with B biallelic loci and individuals are considered incompatible if the number of loci carrying different alleles becomes larger than a fixed threshold G . Genome size, B , has a key role in defining the possible geographical modes of speciation, even when the proportion G/B of allowed genetic incompatibilities is fixed (de Aguiar 2017). And second, the model is ecologically neutral and reproductively isolated species evolve in response to genetic and spatial restrictions imposed on reproduction (de Aguiar et al. 2009).

The incorporation of spatial restriction makes this model particularly convenient to study geographical modes of speciation (for a detailed description of the model, see "Methods" Section). By changing only two model parameters, the degree of restriction to gene flow and the total number of loci, speciation can change from strongly (parapatric) to weakly (sympatric) space dependent (de Aguiar et al. 2009; de Aguiar 2017). For simplicity, we hereafter refer to these two parameters as the size of mating range and the genome size, respectively. In order to effectively compare results generated by genome sizes that differ in many orders of magnitude, we fixed the proportion G/B between threshold value and genome size in all simulations. Therefore, increasing the total number of loci (B) also increases the absolute number of divergent loci recognized as the compatibility threshold, but not the relative proportion of accepted differences. In each case, we computed the corresponding phylogenetic tree from ancestor-descendant relationship information

about extant individuals at the end of the simulation (Costa et al. 2017) and calculated three indexes assessing tree topology: tree balance (Sackin index), branch sizes distribution (γ -statistic), and acceleration of diversification (α -value). We show that the genome size affects both tree balance and rates of diversification through time. The size of the mating range affects only the acceleration of diversification rates. We compare our results with empirical phylogenies focusing on evolutionary radiations, which are a burst in species diversification in which a lineage of species occupies a large (adaptive) or a minimal (non-adaptive) diversity of ecological roles (Rundell and Price 2009). Our findings agree with previous results in that sympatric speciation produces more balanced trees than parapatric speciation (Barraclough and Vogler 2000; Losos and Glor 2003; Pigot et al. 2010), but we expand these results by linking geographic modes with genome size and rates of speciation. The good agreement between the degree of restriction in gene flow in the model and the corresponding patterns seen in the empirical trees of adaptive radiations suggests that the signatures on the tree balance and on the acceleration of the diversification rate are either direct consequences of the geographic mode of speciation or that the similar patterns are generated by adaptive processes not modeled here. Finally, we show that our model of non-adaptive radiations can display slowdown in diversification rate and the overshooting effect (an early increase in number of species followed by a decline that plateaus (Gavrilets and Losos 2009)), patterns usually considered signatures of the adaptive counterparts of the diversification process (Gavrilets and Vose 2005; Gavrilets and Losos 2009).

MATERIALS AND METHODS

Model

The spatial model, we discuss here is a modified version of the model proposed by de Aguiar et al. (2009). We consider a population of $M=1000$ haploid individuals randomly distributed in a square lattice of linear size $L=100$ with periodic boundary conditions (individuals can, by chance, occupy the same lattice site). The choice of haploids is for computational simplicity, since diploid models lead to qualitatively similar results (Schneider et al., 2016). The value of M is kept constant at each time step. Genomes are represented by binary strings of size B , $\{\sigma_1^i, \sigma_2^i, \dots, \sigma_B^i\}$ for individual i , where each locus σ_k^i , can assume the allele values 0 or 1. Here, we refer to B as genome size for simplicity. The genetic distance d between two individuals i and j is the Hamming distance between the corresponding sequences and measures the number of genes bearing different alleles:

$$d^{i,j} = \sum_{k=1}^B |\sigma_k^i - \sigma_k^j|. \quad (1)$$

Mating is restricted by genetic similarity and by spatial proximity, so that an individual i can only choose as mating partner those inside a circular neighborhood of radius S centered in its spatial location (the mating range), and whose genetic distance satisfies $d^{i,j} \leq G$ (Gavrilets et al. 2000). The parameter G represents the genetic threshold of reproductive isolation, by which individuals find potential compatible mates. Therefore, partner choice is determined solely by the compatibility condition $d^{i,j} \leq G$ and not by minimizing the genetic distance $d^{i,j}$. The parameter S represents the size of the mating range. Given the set of compatible individuals within the mating range, the focal individual mates randomly with one of the potential compatible partners. In the simulations, we used $G = 0.05B$, which means that genetic threshold will always be proportional to genome size. The scaling of model parameters with population size was discussed in (Baptestini et al. 2013). The model dynamics depend critically on genome size (de Aguiar 2017), so that for very large genomes speciation may occur in sympatry ($S \rightarrow L$) and also in parapatry ($S \ll L$). However, for smaller genomes, it only happens if mating is restricted to sufficiently close neighbors—parapatry ($S \ll L$). In the limit of infinitely large genomes and $S = L$ the model recovers the Derrida-Higgs dynamics (Higgs and Derrida 1991; de Aguiar 2017).

Each one of the M individuals has a chance of reproducing, but there is a probability Q that it will not do so, accounting for the fact that not all individuals in the present generation will be first parents of the next. In case the focal individual does not reproduce, another one from its mating range is randomly chosen to reproduce in its place, which maintains the population size approximately constant. In our simulations, we set $Q = 0.37 \approx e^{-1}$, which corresponds approximately to the probability that an individual is not selected in M trials with replacement, $(1 - 1/M)^M \approx e^{-1}$, in accordance with the Derrida-Higgs model (Higgs and Derrida 1991; de Aguiar 2017). The first parent (the focal individual or a neighbor) chooses a compatible second parent within its mating range of radius S . The number of individuals in the mating range may be close to zero due to fluctuations in the spatial distribution. To avoid this situation, we follow the procedure introduced in (de Aguiar et al. 2009): if the number of compatible mates in the range is smaller than P ($P = 3$ in our simulations), the individual expands the search radius to $S + 1$. If the number of compatible mates is still smaller than P , the process is repeated up to $S + 2$, and if there is still less than P potential mates, another neighbor is randomly selected to reproduce in its place (de Aguiar et al. 2009).

The reproduction is sexual: the offspring inherits, locus by locus, the allele of either parent with equal probability. The reproductive process is repeated until all M individuals had its chance to reproduce. After reproduction, the resulted offspring is also subjected to a mutation rate μ per locus ($\mu = 0.00025$ in our simulations). This is a modification with respect to the original model

in (de Aguiar et al. 2009), where offspring genomes were generated with a single crossover of the parental genomes. The dynamics is constructed in such a way that offspring is placed close to the location of the original parents and the homogeneous distribution of the population is preserved at all times. In either case, the offspring generated will be positioned exactly at the location of the focal parental individual, or it will disperse with probability D (here we set $D = 0.01$) to one of the 20 nearest neighbors (radius approximately equal to $\sqrt{5} \approx 2.24$). Therefore, close to the location of every individual of the previous generation, there will be an individual in the present generation, keeping the spatial distribution homogeneous and avoiding the formation of spatial inhomogeneities. With this choice, we also avoid a stronger influence of dispersal on gene flow, regarding the latter as a result essentially of mating.

We identify a species as a group of individuals reproductively isolated from all others by the genetic threshold on mating defined by parameter G . Not all members of the group have to be able to mate with each other, but could maintain an indirect gene flow through an intermediary individual. Therefore, if individuals A and B are compatible and so are B and C , but A and C are reproductively isolated, A , B and C will belong to the same species, owing to the ongoing gene flow among all of them. No condition on spatial proximity is imposed on the members of a species.

Individuals are genetically identical at the beginning of the simulation but, as time proceeds, mutations accumulate and reproductively isolated species branch off the population. After a transient time of increase in species number, speciation events are balanced by extinctions and the number of species remains approximately constant. During the transient, extinctions and incomplete speciation events are also observed (Rosindell et al. 2010) (see Supplementary Section 1 available on Dryad at <http://dx.doi.org/10.5061/dryad.4kn5j5d>). In all cases studied here the populations evolved only up to equilibration time, when speciation and extinction balance, so as to describe the process of radiation from a population inhabiting a single area.

In order to evaluate how the size of mating range affects the diversification patterns observed in phylogenetic trees, we vary the parameters S (mating range) and B (genome size) of the model. Fixing a large enough value of genome size, $B = 150,000$, and varying the size of the mating range (S), we control the geographical mode of speciation, ranging from parapatry with low levels of gene flow (small S) to sympatry with high levels of gene flow (large S). The sympatric case is characterized by a value of S that is large enough to allow most individuals in the initial state to potentially interbreed—see “Large Genome Size” in Results. On the other hand, fixing a small value of S , such as $S = 5$, and increasing B from 150 to 150,000, we evaluate the role of genome size in parapatric

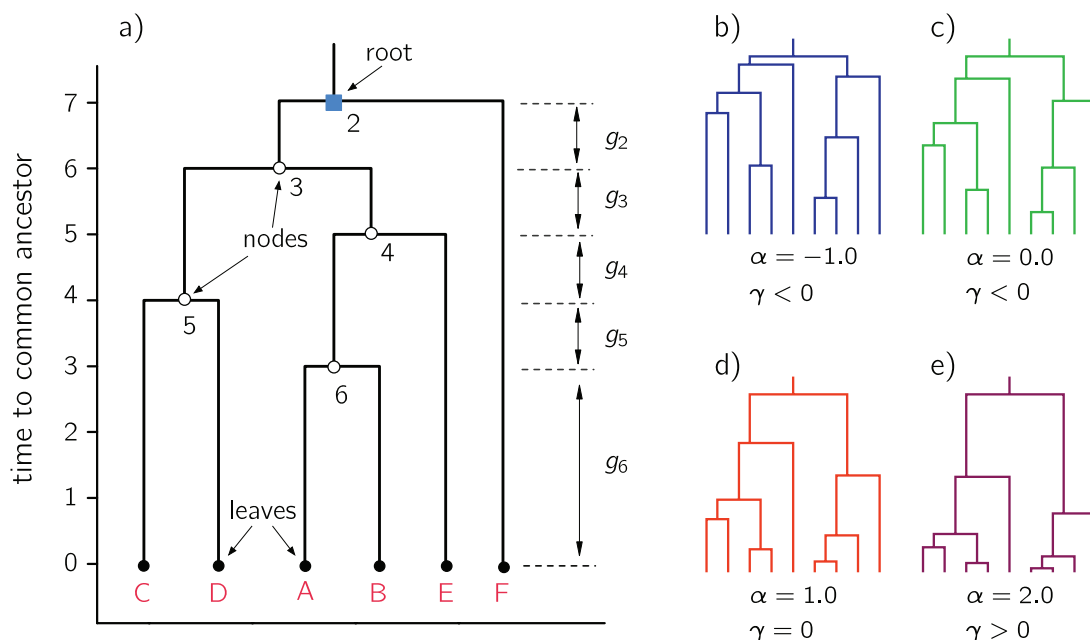


FIGURE 1. (a) Example of a phylogenetic tree with six species represented by filled circles as the leaves of the tree. Open circles denote the nodes, which are numbered from 2 (topmost node, also called the root) to N . Letters below the leaves label the species. Times between speciation events, g_k , are used to calculate the γ -statistic. b–e) Phylogenetic trees showing different distributions of branch length and the relation between statistics γ and α . The tree represented at (c) has a constant bifurcation (speciation) rate per unit of time—Yule model, resulting in constant inter-node distances ($\alpha=0.0$). The tree represented at (d) has a constant bifurcation (speciation) rate per branch—pure-birth model ($\gamma=0.0$).

speciation with low levels of gene flow—see “Small Mating Range” in Results. We have fixed $G/B=0.05$ in all simulations. Analytical and simulation results for the time to divergence for different values of G/B can be found in [Supplementary Section 2](#) available on Dryad. We do not test for small B and large S , because speciation does not occur under this combination of parameters ([de Aguiar 2017](#)).

Rooted Bifurcating Trees and Quantifying Indexes

Differently from distance-based methods usually employed for species relationship inference, here, we constructed the phylogenetic trees based on the ancestor-descendant relationships among individuals. In each generation, we recorded parenthood for the entire population and we registered the time to the most recent common ancestor (MRCA) between each pair of individuals, tracing their genealogical relationships registered at each time step. In this way, at the end of the simulation we had the time to the MRCA between all individuals of the extant population, which evolved from a single ancestor. We define the branch lengths and the structure of the phylogenetic trees using only one individual of each species, since both are insensitive to individuals’ choice in our simulations ([Costa et al. 2017](#)). The most recent common ancestor of all the extant species of a tree was called the tree-MRCA, located in the root node of the tree. Figure 1 describes the most important elements of a rooted bifurcating tree. In order to quantify and compare different trees we use the following indexes:

- Sackin index ($I(N)$)**—The Sackin index evaluates tree imbalance ([Sackin 1972](#); [Blum and François 2005](#); [Frost and Volz 2013](#)) by measuring the distances between the leaves and the root. It is defined as $I(N)=\sum_j d_j$, where d_j is the number of nodes to be traversed between each leaf j and the root including the root ([Dearlove and Frost 2015](#)). The minimum value occurs for the most symmetric bifurcating tree and it can be shown that $\min(I) \approx N \ln(N)/\ln(2)$. This expression is exact when the tree size N is a power of 2, and it is an approximation in the other cases. The maximum value of the Sackin index is found in trees that are most unbalanced and is given by $\max(I)=(N-1)(N+2)/2$. For the tree (a) in Figure 1 $I(6)=18$, whereas the maximum value for 6 leaves is 20 and the minimum 16. For the Yule model, the average Sackin index for a tree with N leaves is given by $E(I(N))=2N \sum_{k=2}^N \frac{1}{k}$. Thus, it is possible to define a relative, normalized, Sackin index as $I_n(N)=\frac{I(N)-E(I(N))}{N}$, which we use in this study. Since $E(I(6))=17.4$ we find $I_n=0.1$ for the tree (a) in Figure 1. We present detailed explanation about the Asymmetric Yule model in [Supplementary Section 3](#) available on Dryad. We used the *sackin* function of the *apTreeshape* package ([Bortolussi et al. 2005](#)) to compute the Sackin’s index on trees and normalize it.
- Alpha value (α)**—The distribution of branch lengths along the tree is usually computed with the

TABLE 1. Empirical radiations. Point in this table is used in the Fig. 5 (“Results” section). N_s is the number of species in each empirical tree.

Point	N_s	Species	Gene flow during diversification	Radiation	References
1	10	Barombi Mbo cichlids	High	Adaptive	Martin et al. (2015)
2	9	Bermin cichlids	High	Adaptive	Martin et al. (2015)
3	35	Hawaiian silversword alliance	Intermediate	Adaptive	Blonder et al. (2016)
4	71	Caribbean <i>Anole</i> lizards	Intermediate	Adaptive	Alföldi et al. (2011)
5	25	<i>Tetragnatha</i> spiders	Intermediate	Adaptive	Gillespie (2004) ^a
6	14	Darwin’s finches	Intermediate	Adaptive	Clarke et al. (2017); Lamichhaney et al. (2015)
7	44	Tanganyika cichlids-1	Low	Adaptive	Meyer et al. (2015)
8	40	Tanganyika cichlids-2	Low	Adaptive	McGee et al. (2016)
9	16	Malawi cichlids	Low	Adaptive	McGee et al. (2016)
10	38	Australian <i>Gehyra</i> geckos	High	Non-adaptive	Heinicke et al. (2011)
11	25	Delphinidae	High	Non-adaptive	Banguera-Hinestroza et al. (2014)
12	74	Philippine sun skinks	High	Non-adaptive	Barley et al. (2013)
13	68	Rodents <i>Rattus</i>	High	Non-adaptive	Rowe et al. (2011)
14	68	Lichen <i>Sticta</i>	Intermediate	Non-adaptive	Simon et al. (2018)
15	46	Caviomorph rodents <i>Ctenomys</i>	Low	Non-adaptive	Álvarez et al. (2017)
16	53	New World titi monkeys	Low	Non-adaptive	Byrne et al. (2016)

^aPersonal communication.

Gamma statistic (γ), defined as (Pybus and Harvey 2000):

$$\gamma = \frac{1}{D} \left[\frac{1}{N-2} \sum_{k=2}^{N-1} T(k) - T(N)/2 \right] \quad (2)$$

where

$$T(k) = \sum_{j=2}^k j g_j; \quad (3)$$

$$D = T(N)/\sqrt{12(N-2)} \quad (4)$$

and g_k is the time interval between speciation events as labeled by the nodes (see Fig. 1). The γ -statistic is constructed in such a way that $\gamma=0$, if $g_k=1/(bk)$. This corresponds to a continuous time process in which all species bifurcate with fixed rate b , leading to $\langle g_k \rangle = 1/(bk)$, $\langle \gamma \rangle = 0$ and $\langle \gamma^2 \rangle = 1$ (see Supplementary Section 4 available on Dryad).

One of the criticisms about γ -statistic is that it depends on the number of leaves in the tree (McPeck 2008; Phillimore and Price 2008). Writing the time between speciation events as $g_k = k^{-\alpha}/b$ we obtain $\gamma = \gamma(N, \alpha)$. This relation can be numerically inverted such that for any tree with given γ and N , a unique α -value can be computed, measuring the changes of speciation rate, i.e., the acceleration of speciation, along the tree. Constant speciation rates per branch corresponds to $\alpha=1$, and constant rates of speciation events at any time corresponds to $\alpha=0$ (trees (d) and (c) in Figure 1 and see Supplementary Section 5 available in Dryad). For the tree in Figure 1(a) $\alpha=-1.58$, which fits into the type described in Figure 1(b).

Empirical Phylogenies

We compared our simulations with 16 known cases of evolutionary (adaptive and non-adaptive) radiations in animals and plants (Rundell and Price 2009; Simoes et al. 2016), listed in Table 1 (check the phylogenetic trees in the Supplementary Section 6 available on Dryad). In order to compare the tree statistics of empirical data with the different degrees of gene flow modeled here (though the parameter S), we define qualitatively the level of gene flow during diversification in empirical data as low, intermediate, or high, based on information of system characteristics as dispersal rate and geographic distribution of species.

Within the set of adaptive radiations (points 1–9), cichlid radiations in crater lakes were considered to have been subjected to the highest levels of gene flow because the lakes are small and these radiations have been traditionally considered examples of sympatric speciation (Martin et al. 2015). Lake Tanganyika and lake Malawi cichlid radiations were considered to have been subjected to the lowest levels of gene flow because the species in these groups inhabit a great lake and are highly phylopatric, exhibiting low levels of dispersal and migration (Kocher 2004). The four terrestrial adaptive radiations were classified as intermediate cases. Studies suggest that in all these cases, islands were initially invaded by a single species and the spreading to different islands gave rise to independent small radiations (Rundell and Price 2009). The analyzed trees encompass all species distributed across these island systems and evolutionary patterns result from the combination between processes occurring between islands with processes occurring within islands. Cichlid radiations in large lakes probably have a comparable history of mixed levels of gene flow (Meier et al. 2017a, 2017b). Nonetheless, given the dispersal potential of these species in relation to the size of lakes Malawi and Tanganyika (orders of magnitude larger than crater lakes), the general assumption that gene flow

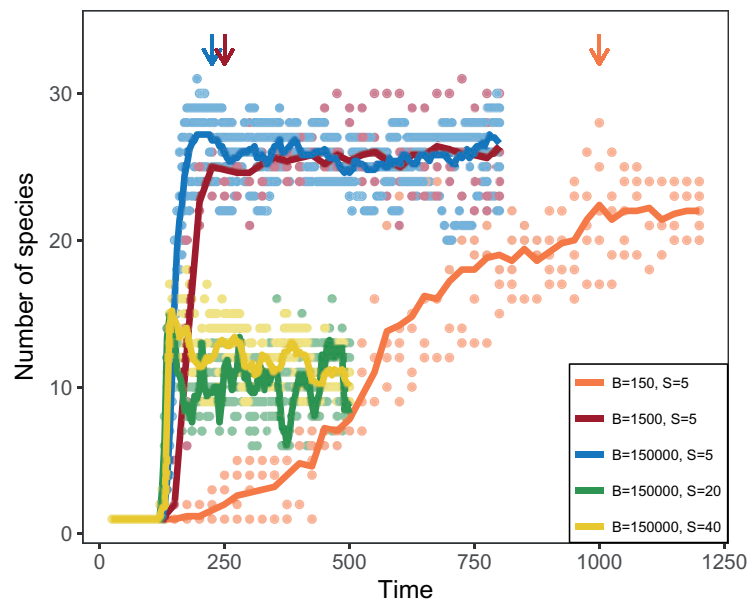


FIGURE 2. Number of species as a function of time for different combinations of mating range, S and genome size, B . Points show results of five realizations for each value of B at each time step, darker points depict overlapping values. Solid lines show the average values. Arrows indicate the simulation time when phylogenetic trees were computed for $B=1,50,000$ (225 generations, blue, all values of S), $B=1,500$ (250 generations, red) and $B=150$ (1,000 generations, orange). For $B=150,000$ and $S=20$ and $S=40$ the number of species reaches a maximum (about 10 species) before equilibration (see [Supplementary Fig. S10](#) available on Dryad). The largest number of species occur for $S=5$, $B=150,000$ and $B=15,000$ (both about 25 species) and $B=150$ (about 20 species).

between lineages has been lower during diversification is warranted.

Regarding the non-adaptive radiations (points 10–16), the Australian geckos, delphinids, Philippine sun skinks, and rodents of genus *Rattus* were considered to have high gene flow during diversification due to the high dispersal rates and/or broadly sympatric distributions observed in these groups ([Heinicke et al. 2011](#); [Banguera-Hinestroza et al. 2014](#); [Barley et al. 2013](#); [Rowe et al. 2011](#)). The lichen of genus *Sticta*, as the terrestrial adaptive radiations, was classified with intermediate gene flow, because also in this system the radiation was initiated by a single long-distance dispersal event (in the Madagascar and Indian Ocean Islands region in this case) followed by multiple dispersal events between islands ([Simon et al. 2018](#)). Finally, the caviomorph rodents and the titi monkeys were considered to have low gene flow during diversification, since the first are strictly territorial and both present restricted dispersal abilities due to their subterranean habits and geographic isolation by rivers, respectively ([Álvarez et al. 2017](#); [Byrne et al. 2016](#)). We considered these as non-adaptive radiations because they display conserved ecomorphological traits, even though the initial phase of radiation presents a burst in speciation rate, traditionally associated with adaptive radiations (but see [Moen and Morlon, 2014](#)).

RESULTS

Gene flow has a strong negative effect on final species richness. Indeed, species richness at equilibration is higher when gene flow is restricted and the number of loci is large (small S and large B , blue and red

curves in Fig. 2), thus facilitating the formation of new species both from the spatial and genetic aspects. Genome size drastically affects the time for equilibration of species number (Fig. 2). For small genomes ($B=150$), equilibration takes about 1,000 generations, and speciation events occur at an approximately constant rate (see also [Supplementary Fig. S12](#) available in Dryad). For large genomes ($B=1,50,000$ and $B=1,500$), equilibration happens earlier, about 250 generations, and diversification patterns resemble those of adaptive radiations, where many species arise early and species accumulation declines over time (see [Supplementary Fig. S13](#) available on Dryad). For large genomes, and intermediate to large mating range ($B=150,000$ and $S=20$ or $S=40$), there is an overshooting effect ([Cavrilets and Vose 2005](#)) and number of species at equilibrium is lower than the maximum reached during the radiation (Fig. 2 and [Supplementary Fig. S10](#) available on Dryad).

Genome and compatibility threshold size have important effects on the radiation patterns. We recall that results are shown for $G/B=0.05$, so that small (large) genome size implies small (large) compatibility threshold. Radiation is slower for small genomes (Fig. 2) with small species branching off the original population and rarely suffering speciation again (see [Supplementary Fig. S12](#) available on Dryad) resulting in unbalanced trees. For large genome sizes, on the other hand, the radiation process is fast, with more abundant species branching off and splitting again before equilibration (see [Supplementary Fig. S13](#) available on Dryad), giving rise to balanced trees. Stochastic fluctuations in mating choice cause the number of individuals per species to change over time,

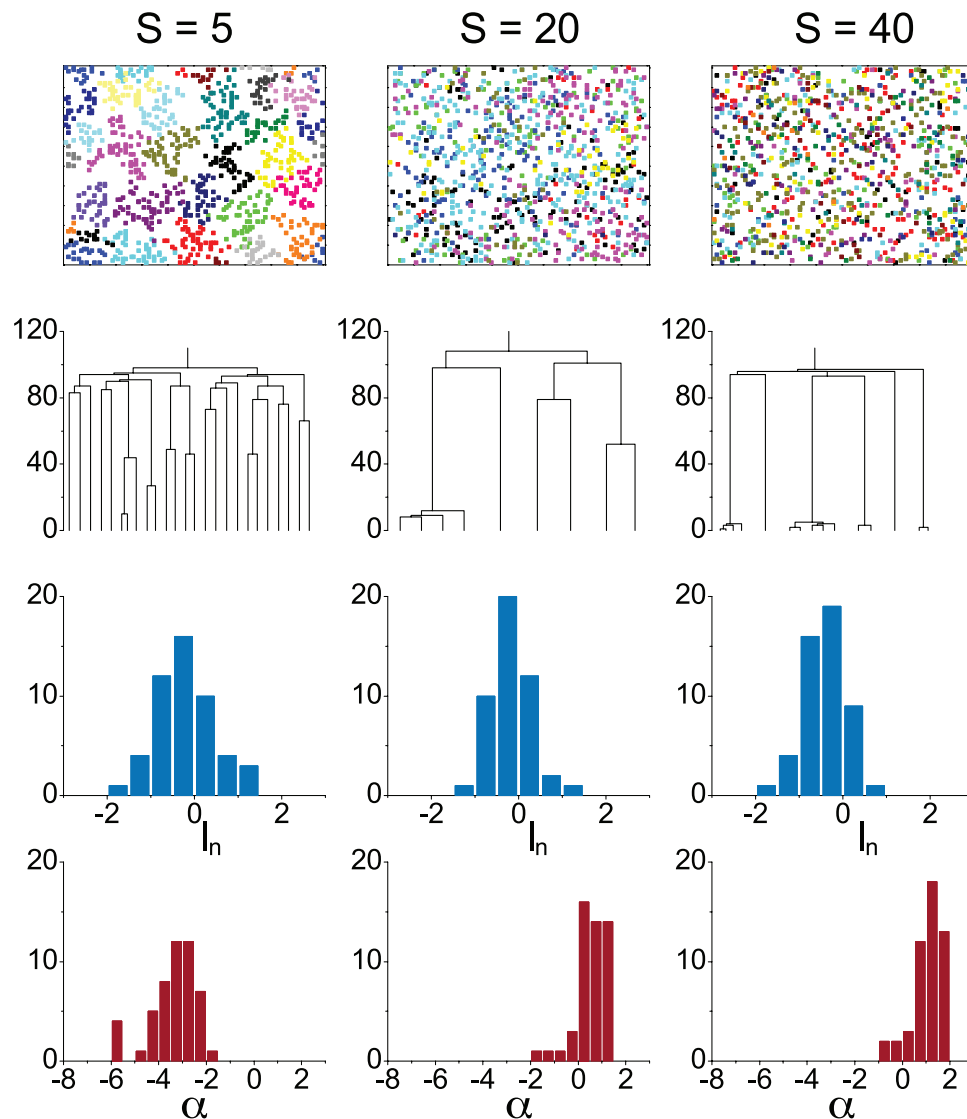


FIGURE 3. Simulations for $B=150,000$ and different values of S . From top to bottom rows: spatial distribution of the populations (different colors - electronic version - or shades of gray - printed version - show different species); examples of corresponding phylogenetic trees; normalized Sackin index distribution (I_n); α -value distribution (α). The distributions were computed for 50 realizations of the process.

eventually leading species to extinction by ecological drift. It is worth noticing that extinctions are negligible during the radiation for large genome/threshold ($B=150,000$, $G=7,500$) and occur at a constant rate for small genome/threshold ($B=150$, $G=7$) (see [Supplementary Figs. S1 and S2](#) available on Dryad). The extinction pattern, however, does not appear to be a consequence of the average abundance distribution of individuals per species, as species have a larger population size for small genome ($B=150$) than for large genome ($B=150,000$) at equilibration time under the same mating range (see [Supplementary Section 8](#) available on Dryad). In [Supplementary Section 9](#) available on Dryad, we show that model parameters S and B can be inferred from the phylogenies with some accuracy. The power of inference of the model has limitations regarding the parameter space covered, the variance of the distributions (size of ellipses) and the number of simulations.

Large Genome

Figure 3 shows the effect of the geography of speciation for large genome size ($B=150,000$, $G=7,500$) with typical species distributions in space (first row) and the corresponding phylogenetic trees (second row). Columns correspond to increasing values of S , i.e., low ($S=5$), intermediate ($S=20$), and high ($S=40$) levels of gene flow during diversification. Gene flow has a key role in the spatial organization of the populations, going from clustered to totally mixed in space. For small S , speciation is preceded by the accumulation of local genetic variation within species. As a result, genetic and physical spaces become coupled by the dynamics and species form in well-defined clusters. As S increases this correlation is lost and species mix in space (see first row in Fig. 3 from left to right). The time to the most recent common ancestor of all the

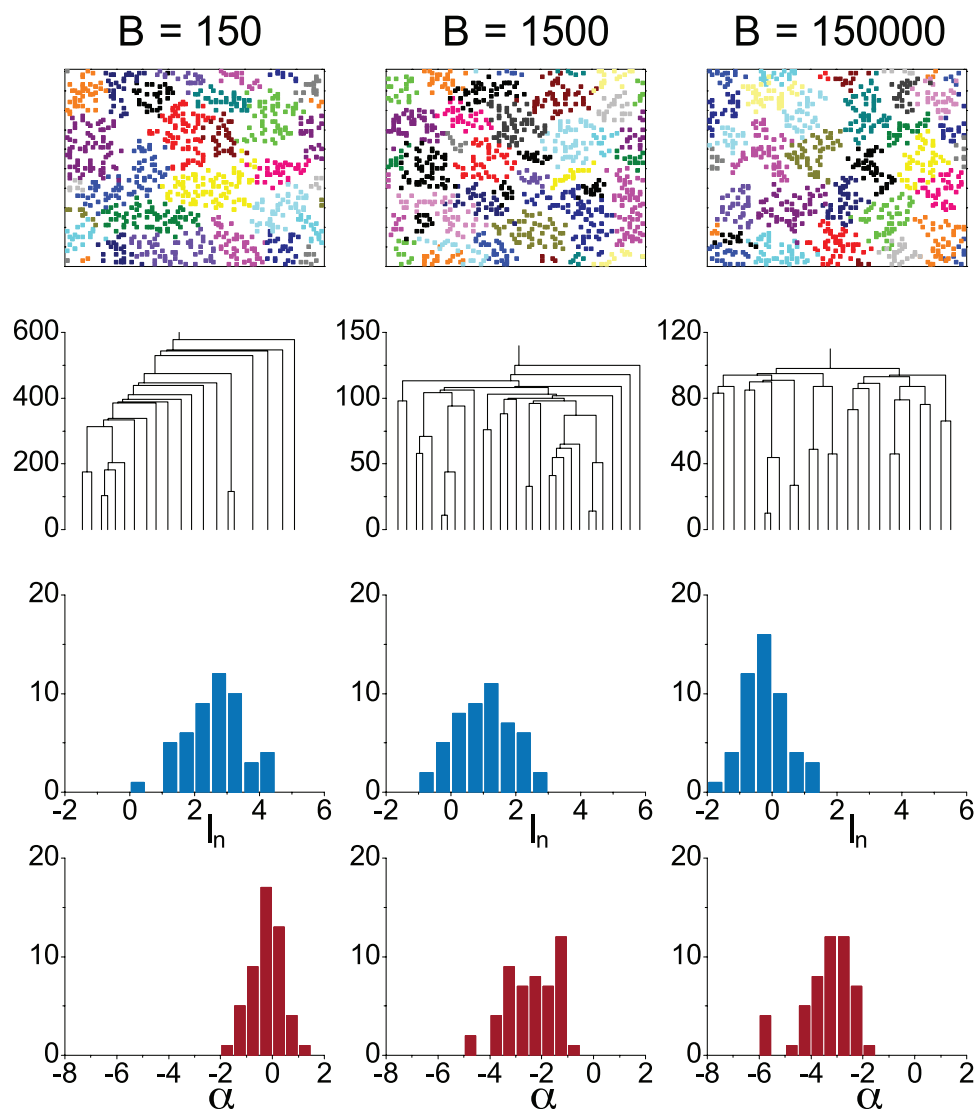


FIGURE 4. Simulations for $S=5$ and different values of B . From top to bottom rows: spatial distribution of the populations (different colors - electronic version - or shades of gray - printed version - show different species); examples of corresponding phylogenetic trees; normalized Sackin index distribution (I_n); α -value distribution (α). The distributions were computed for 50 realizations of the process.

extant species of the tree (tree-MRCA) is similar for all values of S (see the time of first bifurcations in the phylogenies in Fig. 3), which is around 100 generations. The last two rows show the distribution of normalized Sackin index and α -value. The normalized Sackin index distributions (third row) are largely insensitive to S and indicate balanced trees, as predicted by the Yule model (Yule 1925). Distribution centers and standard deviations are -0.1 ± 0.6 , -0.2 ± 0.4 , and -0.4 ± 0.5 for $S=5$, 20, and 40 respectively. In Supplementary Fig. S7 available on Dryad, we superimposed the simulated histograms with the predictions of an asymmetric Yule model (see Supplementary Section 3 available on Dryad for more information). Alternatively, the acceleration of the diversification rate, here characterized by the α -value, is highly influenced by gene flow: with stemmy trees when gene flow is restricted and tipy trees when

gene flow is high (see second row in Fig. 3). Distribution centers and standard deviation of α -values are: -3.4 ± 0.9 ($S=5$), 0.6 ± 0.6 ($S=20$), and 1.0 ± 0.7 ($S=40$).

Small Mating Range

Figure 4 shows the effect of genome size (for fixed $G/B=0.05$) on phylogenetic trees when speciation is strongly dependent on space (parapatry, $S=5$) with spatial distributions of species (first row), and the corresponding phylogenetic trees (second row). Now, columns correspond to increasing values of genome size (B) and compatibility threshold (G). In all cases, species display strong correlation between genetic and physical spaces, as expected for a small S . However, the time to the most recent common ancestor of all the extant species

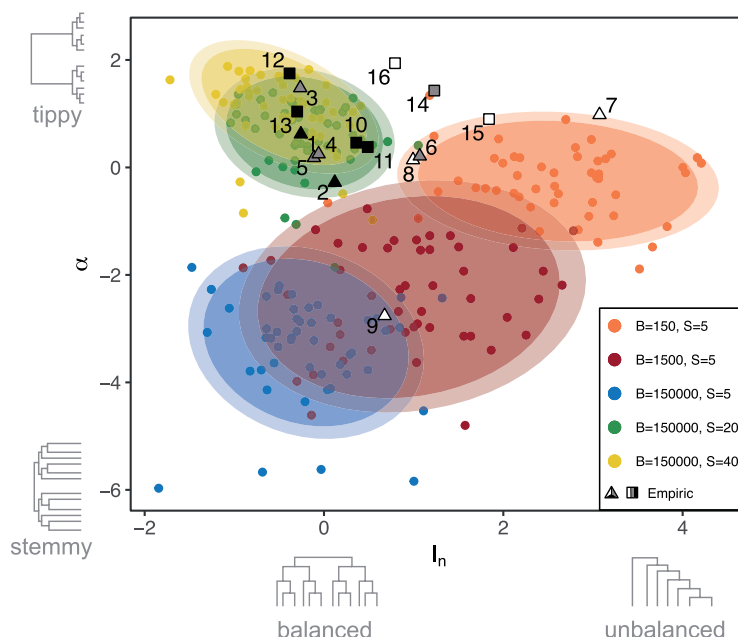


FIGURE 5. Distribution of α -value and normalized Sackin index I_n for different model parameters. Populations evolved up to species number equilibration time (50 replicates). Ellipses comprehend 90% (darker) and 95% (lighter) of the simulated replicates for each parameter combination. Numbered triangle and square symbols represent phylogenetic tree indexes estimated from empirical data as listed on the Table 1 in the “Materials and Methods” section. Triangles represent adaptive and squares represent non-adaptive radiations. The colors of triangles and squares are related to the gene flow during diversification as defined on Table 1: white = low, gray = intermediate, black = high. The two upper overlapping ellipses correspond to $B=150,000$: $S=20$ (lower, green) and $S=40$ (upper, yellow). The three remaining ellipses, in the diagonal of the plot, from upper right to lower left are for $S=5$: $B=150$ (orange), $B=1,500$ (red) and $B=150,000$ (blue).

of the tree (tree-MRCA) depends critically on B and G . For $B=150$, $G=7$ the time to the tree-MRCA is around 600 generations (see the time of first bifurcation in the left phylogeny in Fig. 4), which differs from the time species started to be formed (around 200 generations, see the time the orange curve detaches from the time axis on Fig. 2), i.e., species formed at the initial radiation (between 200 and 600 generations) went extinct. As B increases, the time of the first speciation event decreases and the equilibration time occurs earlier (Fig. 2).

The distribution of normalized Sackin index (third row) are highly unbalanced for small B ($I_n = 2.6 \pm 0.9$ for $B=150$, $G=7$). As B increases the center of the distribution shifts monotonically to the left: $I_n = 1.0 \pm 0.9$ for $B=1,500$, $G=75$ and $I_n = -0.1 \pm 0.6$ for $B=150,000$, $G=7,500$. In Supplementary Fig. S7 available on Dryad, we superimposed the predictions of the asymmetric Yule model for the histograms, with $\delta=0.3$ for $B=150$, $G=7$, $\delta=0.65$ for $B=1,500$, $G=75$, and $\delta=0.8$ for $B=150,000$, $G=7,500$. The α -value distributions (last row) also display strong dependence on genome/threshold size, with speciation rate decelerating as genome size increases ($\alpha = -0.3 \pm 0.6$ for $B=150$, $G=7$, $\alpha = -2.4 \pm 1.0$ for $B=1,500$, $G=75$, and $\alpha = -3.4 \pm 0.9$ for $B=150,000$, $G=7,500$).

Comparison with Empirical Data

Figure 5 shows how the acceleration of diversification (α -value) and tree balance (normalized Sackin index, I_n)

are influenced by mating range and genome size in the simulated model (colored dots and ellipses). The mating range affects specially the acceleration of diversification ($F=508.18$, $\eta^2=0.87$), but the balance of the trees are only marginally affected by this parameter ($F=3.39$, $\eta^2=0.04$). Genome size, for fixed G/B , affects both the acceleration of diversification ($F=126.08$, $\eta^2=0.70$) and tree balance ($F=151.53$, $\eta^2=0.67$). The numbered symbols in Figure 5 show indexes of phylogenetic trees estimated from the empirical radiations listed in Table 1 (see also the phylogenetic trees in Supplementary Fig. S11(a–p) available on Dryad).

Most empirical points fall in two regions, corresponding to: (i) $B=150,000$ (large genome size) with $S=20$ or $S=40$ (moderate to high gene flow) and balanced trees—green and yellow regions (1, 2, 3, 4, 5, 10, 11, 12, and 13) and (ii) $B=150$ (small genome size) with $S=5$ (low gene flow) and unbalanced trees—orange ellipse (points 6, 7, 8, and 15). The Malawi cichlids (point 9) fall in the region of low gene flow with large genome. The lichen *Sticta* and the New World titi monkeys (points 14 and 16) fall between the low/high gene flow and small/large genome size regions (orange/green-yellow ellipses). The α -value close to 1, corresponding to constant speciation rate per branch, was characteristic of speciation with moderate or high levels of gene flow ($S=20$ and 40, sympatry). Simulations of low levels of gene flow ($S=5$, parapatry), on the other hand, displayed average values of α -value

smaller or equal to 0, associated with a slow down of diversification rate.

DISCUSSION

In this article, we investigated how different geographical modes of speciation leave signatures on phylogenetic trees using a neutral spatially-explicit IBM. The origin of new species resulted from spatial and genetic thresholds, denoting sympatric and parapatric speciation under certain parameter values. Phylogenies were constructed by tracking ancestor-descendant relationships between individuals (Costa et al. 2017). We simulated evolutionary radiations and calculated two indexes assessing tree topology: tree balance (Sackin index) and acceleration of diversification (α -value). We compared the indexes obtained from simulated results with those found in trees estimated from empirical data of adaptive and non-adaptive radiations. We found signatures of the geographical mode of speciation displayed as the relationship between the acceleration of diversification (y -axis in Fig. 5) and the balance of the trees (x -axis in Fig. 5). Adaptive and non-adaptive empirical radiations exhibit macroevolutionary patterns that are consistent with gene flow simulated for the neutral model (squares and triangle symbols in Fig. 5).

Evolutionary radiations are often thought to be characterized by either a slowing down in diversification rates, associated with ecological speciation and the process of niche filling (negative diversity-dependence) (Phillimore and Price 2008; Rabosky and Lovette 2008; Gavrilets and Losos 2009; Gascuel et al. 2015), or by a speeding up in diversification associated with coexistence of newly branched species driven by ecological interactions (positive diversity-dependence often referred as “diversity begets diversity”) (Emerson and Kolm 2005; McPeck 2008; Gascuel et al. 2015; Burin et al. 2018). In our model, the acceleration of diversification (α -value) is an important measure of the overall speeding-up or slowing down of speciation rate during the radiation (see [Supplementary Fig. S9](#) available on Dryad). We specifically refer to speciation rates because extinctions are either irrelevant (for large genomes) or constant (for small genomes) during the radiation (see [Supplementary Section 1](#) available in Dryad). Speciation rate has an accelerated initial phase followed by a slowing down before equilibration, which results in an S-shaped curve of number of species through time (Fig. 2 and [Supplementary Fig. S10](#) available in Dryad). For small genomes, however, this pattern is attenuated. The geography of the diversification process affects diversification slowdown (Moen and Morlon 2014; Pigot et al. 2010). Stable geographic ranges have been associated with slowdown in diversification rates (Pigot et al. 2010), which in our model encompasses parapatric radiations with large genomes and resulting in slowdown of diversification rates (negative α -value). Alternatively, sped up diversification rates are characteristic of sympatric radiations with large genome or parapatric

radiations with small genome (positive α -value). We emphasize that we have fixed the ratio between threshold value and genome size (G/B) in all simulations, so that small(large) genome sizes always implies small(large) compatibility threshold. Peripatric speciation produces highly unbalanced trees (Pigot et al. 2010), which, here, were associated with parapatric speciation (positive I_n). The distinction between sympatric and parapatric radiations with nearly constant diversification rates is in the tree balance. We reinforce the conjecture that when considering diversification alone, similar patterns of diversification rate in time can result from different processes (McPeck 2008; Yoder et al. 2010). In our results this is reflected in the size of the ellipses representing the different scenarios simulated here. The stochastic nature of the model implies that different trees can be generated with the same parameter values and, more importantly, different parameters can produce similar trees. Although the “low resolution” manifested by the size of the ellipses makes the connection between trees and processes not so accurate, sympatry, represented by the green and yellow ellipses (balanced and tippy trees) can still be clearly distinguished in the two-dimensional tree topology space from parapatry, blue and orange ellipses for large and small genomes/thresholds, going from balanced stemmy to unbalanced tippy (Lewitus and Morlon 2016).

The spatial scale relevant to the speciation process depends on the intensity of gene flow, so that species with lower levels of gene flow require smaller areas to speciate (Kisel and Barraclough 2010). In island radiations, reduced gene flow increases the probability of speciation (Kisel and Barraclough 2010). We report a similar pattern, as for reduced gene flow (low S), speciation can occur regardless of the genome/threshold size considered. In models of adaptive speciation, however, the source of genetic divergence emerges from competition and gene flow promotes speciation (Dieckmann and Doebeli 1999). The connection between the spatial component of the speciation process and the degree of gene flow brings up the question of how the underlying process driving genetic divergence affects diversification in the macroevolutionary scale. In our case, genetic divergence evolves as a consequence of the emergence of reproductive isolation, arising directly from microevolutionary processes incorporated in the model, and the degree of gene flow can be directly related to the size of the mating range and, consequently, to the geography of speciation. When the genome is large, mutations spread over loci uniformly, so that the average genetic distance between any two individuals is about the same and increases monotonically with time (see [Supplementary Section 2](#) available on Dryad). When the average distance reaches the threshold G the population splits very quickly into several species (see [Supplementary Section 7](#) available on Dryad) giving rise to stemmy trees. However, because there is potential gene flow between all individuals until very close to speciation, species form with relatively large populations and split again later, producing symmetric

trees. For small genomes speciation is slower, mutations accumulate closer to each other and species arise one by one from the main population, giving rise to asymmetric trees (Fig. 4). Mating range affects gene flow directly, so that large values of S slows down speciation, leading to tippy trees. Because speciation for large S only occurs for large B , the trees will also be symmetric.

An additional component contributing to the evolution of genetic divergence in our model is related to the combined effect of genome and compatibility threshold size with large values facilitating speciation. Interestingly, this is also a dynamic feature found in adaptive evolution models (Flaxman et al. 2014). Studies investigating the relationship between genome size and diversification rates using empirical data show that, in most cases, small-genome taxa present higher diversification rates across different species groups (Kraaijeveld 2010). This relation is grounded in the assumption that the smaller the genome, the quicker the genomic rearrangements, and the greater the rate in which mutations can accumulate (Kraaijeveld 2010). This body of assumptions are not met by our modeling approach, as we do not incorporate genomic rearrangements and their consequences to organism function. In our model, mutations accumulate at the same rate despite genome size and population divergence appears to be faster as genome size increases. In fact, in angiosperms, there is strong evidence for a positive relation between the rate of genome size evolution and speciation rate, where genome size evolution is mainly related to genome duplications (as polyploidy) (Puttick et al. 2015). Also contributing to the debate between genome size and diversification rates, and in agreement with our results, actinopterygian fishes present a positive correlation between genome size and species richness among genera (Mank and Avise 2006). Even though there are evidences supporting the relationship between genome size and rate of diversification, we do recognize they are not striking and the debate is still open (Kraaijeveld 2010). Additionally, we are aware that genome size or the number of genes involved in speciation are parameters hard to correlate and estimate in empirical systems. The signatures left by the effect of genome size on the speciation process may be smaller or greater than the actual measurable genome metrics (for instance, C-value and G-value). This effect can be even more pronounced given non-linearities of genetic architecture. The part of the genome under selection, the amount of non-coding DNA, genetic interactions (pleiotropy and epistasis) and splicing processes can alter the effective size of the genome (Taft et al. 2007; Sanjuán and Elena 2006; Wagner et al. 2007; Brett et al. 2002).

Empirical data display structural phylogenetic patterns that can be associated with different geographic modes of speciation in our model. Most of the data presented here have tree balance and speed of diversification statistics falling into model predicted regions that are in accordance with the hypothesized

degree of gene flow during the radiation of these systems (Table 1 and Fig. 5). Groups with intermediate gene flow (Hawaiian silversword alliance, *Anole* lizards, *Tetragnatha* spiders, Darwins finches, and lichen *Sticta*—points 3, 4, 5, 6 and 14, respectively) were located in varied regions, with some falling into either sympatric (points 3, 4, and 5) or parapatric (point 6) regions, and one (point 14) located in the transition between these regions. This variation could be attributed to the myriad of factors promoting speciation in these groups as they all belong to island systems, in which radiations are especially prone to occur with parapatric and sympatric components playing preeminent roles during diversification (Rundell and Price 2009; Simoes et al. 2016). The phylogenetic tree of the New World titi monkeys (point 15), hypothesized to have radiated with low gene flow, does not resemble the trees generated with low gene flow by our model, which might signalize that rivers are not effective barriers to gene flow as previously suggested for the Amazonian monkey groups (Santorelli et al. 2018). Most of empirical trees are tippy, indicating that this pattern might not only be attributed to non-ecological modes of speciation but could also be a prevailing pattern in evolutionary radiations independently of the nature of processes driving diversification (McPeck 2008).

The empirical adaptive (non-neutral) radiations also fell in structural phylogenetic regions simulated by our neutral model. This suggests that the geographic mode of speciation can leave signatures in the balance and acceleration of diversification of phylogenetic trees, irrespective of the process being adaptive or non-adaptive. On the other hand, it is also possible that adaptive processes generate phylogenetic patterns overlapping with those predicted by our neutral model. Further studies including adaptive processes are needed to resolve this matter. A prolonged geographical isolation in the early history of a system resulting in multiple reproductive isolated species, characterizing a non-adaptive radiation, could also be confounded with an adaptive one, as species may diversify ecologically when conditions change (Rundell and Price 2009). Even in cases traditionally considered as adaptive radiations, such as Darwins finches and African cichlids, previous studies (Grant and Grant 2011; Losos and Ricklefs 2009; Seehausen 2015; Moen and Morlon 2014; Simoes et al. 2016) show that geographical processes seem to be an important promoter of diversification patterns along with competition for resources and niche filling. Other examples of this misdiagnose can be found in systems with many species within a radiating clade with allopatric distributions, such as the *Ctenomys* caviomorph rodents, *Anoles* lizards, *Tetragnatha* spiders, Darwin's finches, and Hawaiian silversword alliance displayed here (Simoes et al. 2016). Although some macroevolutionary patterns—such as an early burst in diversification and overshooting effect—have been considered signatures of adaptive radiations (i.e., fueled by adaptation to distinct niches), there

is a debate about whether non-ecological processes may result in similar patterns (McPeck 2008; Yoder et al. 2010; Gascuel et al. 2015). These findings might explain the matching between the patterns observed in empirical adaptive radiations and our simulated neutral radiations, although only the inclusion of non-neutral processes in the model (as in Gascuel et al., 2015) would provide clearer interpretations.

Observed association between ecological divergence and diversification does not necessarily indicate that speciation was caused by adaptation to different environments, because ecological differences can evolve after diversification (Rundell and Price 2009). Moreover, niche conservatism followed by allopatric divergence may lead to similar patterns of early diversification (Rundell and Price 2009). Other non-ecological models of rapid radiations have predicted dispersal limitation and isolation of species without environmental differentiation as potential drivers of rapid diversification (Wiens 2004; Rundell and Price 2009). Factors associated with non-adaptive (neutral) and adaptive (non-neutral) radiations could operate synergistically (Simoës et al. 2016), being difficult in empirical studies to isolate the relative importance of each process through the phylogenetic summary statistics employed here. The *Tetragnatha* spiders are an illustration of this mixture of processes, as they contain strongly ecologically differentiated sympatric species, as well as ecologically similar allopatric or parapatric species (Gillespie 2004; Rundell and Price 2009). Additionally, some signatures could be left during the radiation specifically by adaptive process that might be better detected by other measures of phylogenetic structure such as phylogenetic diversity.

Our results contribute to bridge the gap between microevolutionary processes and macroevolutionary patterns. We incorporated a broad range of speciation mechanisms in which the microevolutionary forces at the individual level such as gene flow and genomic architecture, play important roles in the diversification. We have shown that speciation based on genetic and spatial restrictions imposed on reproduction predict clear macroevolutionary patterns in phylogenetic trees. Our neutral model recovers a wide range of patterns associated to evolutionary radiations including the speeding up, followed by the slowing down of diversification rates and even the overshooting effect, which are considered a signature of adaptive radiations (Gavrillets and Losos 2009). Our findings also corroborate previous results in that the geographic mode of speciation is important to the tree balance: sympatric speciation produces more balanced trees than parapatric speciation (Barracough and Vogler 2000; Losos and Glor 2003; Pigot et al. 2010). We have shown these effects for neutral simulated radiations and observed similar patterns in both adaptive and non-adaptive empirical sets. We hypothesize that the observed signatures in Sackin index and α -value in evolutionary radiations are possibly related to gene flow and genome size. Confirmation of this hypothesis would require further investigations with models of adaptive radiations.

We further hypothesize that adaptive/ecological components leave signatures in other phylogenetic tree statistics. For instance phylogenetic diversity (Faith's PD), and the biogeographically weighted evolutionary distinctiveness (BED), (Tucker and Cadotte 2013) might summarize more accurately the role played by the adaptive components of evolutionary radiations. Expanding our analysis to include adaptive speciation processes will be important to improve our understanding on the phylogenetic signatures. Even within the framework of our model other microevolutionary features, such as population density (M/L^2), compatibility threshold (G/B), dispersal probability, evolution of genome size and mating range (variation of B and S) and mutation rate (μ) can be related to realistic features and might also be explored to understand the importance of other microevolutionary processes on the macroevolutionary patterns.

SUPPLEMENTARY MATERIAL

Data available from the Dryad Digital Repository: <http://dx.doi.org/10.5061/dryad.4kn5j5d>.

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